

VGChartz 2024 — Video Game Sales Analytics

64,016 Game Records | 81 Consoles | 20 Genres

Source: [VGChartz.com](https://vgchartz.com) | Tools: Excel · Power BI · DAX

Prepared: May 2026

1. Project Overview

This project analyzes the VGChartz 2024 video game sales dataset — comprising 64,016 game records across 81 platforms, 20 genres, and hundreds of publishers — to uncover business-relevant insights about the global gaming industry. The analysis spans regional sales performance across North America, Japan, PAL regions, and Rest of World, examining genre trends, publisher dominance, console lifecycle patterns, and the correlation between critic review scores and commercial success. The final deliverable is an interactive Power BI dashboard supported by a fully documented analytics pipeline.

- What is the project about?** The project analyzes structured sales data from VGChartz — a leading video game tracking database — to understand the commercial performance of 64,016 games released across 81 gaming consoles from 1975 to 2024. It identifies which titles, genres, publishers, and platforms dominate global and regional markets.
- What problem are you solving?** The video game industry is highly competitive and regionally fragmented. Publishers and platform holders lack a consolidated, visual view of which genres and consoles drive revenue, how critic scores influence sales, and which markets hold the highest commercial opportunity. This project solves that by transforming raw, incomplete data into actionable intelligence.
- Why is this analysis important?** Decisions around game launches, marketing spend, platform exclusivity, and regional pricing directly impact revenue. Data-driven insights from this analysis enable publishers, developers, and platform holders to allocate resources more effectively — targeting the right genre, on the right platform, in the right market, at the right time.

2. Objectives

The key goals of this project are:

- Identify key sales trends and patterns across genres, consoles, and release eras.
- Analyze publisher and developer performance to determine global and regional market leaders.
- Examine the relationship between critic scores and total commercial sales figures.
- Compare regional sales distributions across North America, Japan, PAL (Europe/Australia), and Others.
- Build an interactive Power BI dashboard covering all objectives with slicers, drill-downs, and bookmarks.
- Derive SMART insights using all four types of analytics: Descriptive, Diagnostic, Predictive, and Prescriptive.

3. Data Sources

Attribute	Detail
Dataset Name	VGChartz 2024 — Video Game Sales
Dataset Source	VGChartz.com (Public, web-scraped database)
File Format	CSV (vgchartz-2024.csv)
Timeline	1975 – 2024 (release dates of games)
Total Records	64,016 game titles
Total Columns	14 columns (12 analytical + 2 dropped metadata)

Attribute	Detail
Key Dimensions	title, console, genre, publisher, developer, release_date
Key Measures	total_sales, na_sales, jp_sales, pal_sales, other_sales, critic_score
Dropped Columns	img (box art URL), last_update (72.1% null — no analytical value)
Cleaned Dataset Size	12,637 records (Blockbuster + Hit tier titles with sales data)

4. Data Cleaning (Preprocessing)

The following preprocessing steps were performed to ensure data quality before analysis:

Missing Value Treatment

Column	Null Count	Null %	Treatment Applied
critic_score	57,338	89.5%	Kept null — filtered out in score-based visuals only
total_sales	45,094	70.4%	Replaced null with 0 (Transform → Replace Values)
na_sales	51,379	80.3%	Replaced null with 0
jp_sales	57,290	89.5%	Replaced null with 0
pal_sales	~51,000	~79.7%	Replaced null with 0
other_sales	48,888	76.4%	Replaced null with 0
release_date	7,051	11.0%	Kept null — excluded from date-based visuals
developer	17	0.03%	Replaced null with "Unknown"

Additional Cleaning Steps

- Removed columns with no analytical value: img and last_update dropped via Right-click → Remove Column.
- Removed duplicate records: Select All Columns → Remove Rows → Remove Duplicates.
- Corrected data types: sales columns set to Decimal Number; release_date parsed using locale (DD/MM/YYYY); text columns set to Text type.
- Renamed columns for readability: na_sales → NA_Sales_M, jp_sales → JP_Sales_M, pal_sales → PAL_Sales_M, total_sales → Total_Sales_M, critic_score → Critic_Score, release_date → Release_Date, title → Game_Title.
- Verified uniqueness: title + console combination checked as unique identifier per record.

Tools Used

- Power BI (Power Query) — all transformation steps: null handling, type fixes, column removal, renaming.

5. Data Transformation

To enhance analysis, the following transformations were applied after data cleaning:

Calculated Columns (Power Query)

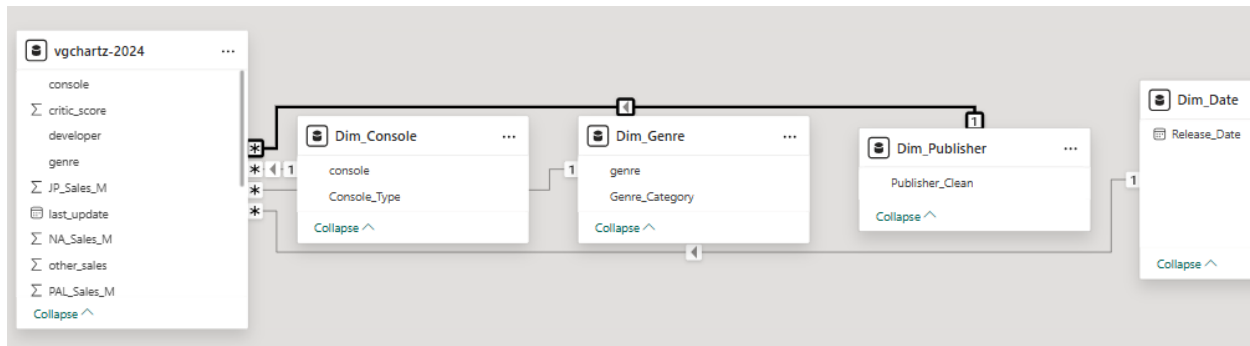
Column Name	Purpose
Release_Year	Enables time-series and year-over-year analysis
Release_Decade	Groups titles into generational eras for era comparison
Sales_Tier	Classifies titles by commercial success level
Score_Band	Segments titles by critic review quality for comparison
Dominant_Region	Identifies primary sales market for each title

DAX Measures (Power BI)

Measure Name	Used In
Total Sales (M)	KPI Cards, all visuals
Total NA / JP / PAL / Other Sales	Regional KPI Cards, Donut Chart
Avg Critic Score	Scatter Plot
Total Games	KPI Card
Sales per Title	Benchmarking Card
NA / JP / PAL Sales %	Donut Chart, KPI Cards

Dimension Tables (Star Schema)

- Dim_Console — unique console list with Console_Type column (Handheld / Home / PC / Other).
- Dim_Genre — unique genre list with Genre_Category column (Action-oriented / Casual / Strategy / Sports).
- Dim_Publisher — unique publishers; blank values replaced.
- Dim_Date — date table built using CALENDAR() from min to max Release_Year; marked as Date Table in Power BI.



6. Exploratory Data Analysis (EDA) & Visualization

Statistical Analysis Performed

- Computed total global sales: 6,605.91 million units across 17,570 records with non-null sales data.
- Computed regional breakdown: NA 3,345.52M (50.6%) | PAL 1,916.83M (29.0%) | JP 687.94M (10.4%) | Other 651.12M (9.9%).
- Identified top publisher: Activision at 722.77M units; top genre: Sports at 1,187.51M units; top console: PS2 at 1,027.76M units.
- Average critic score across 6,678 scored titles: 7.22 / 10.
- Sales tier distribution in cleaned dataset: 91 Blockbuster titles (>5M units); 909 Hit titles (1–5M units).

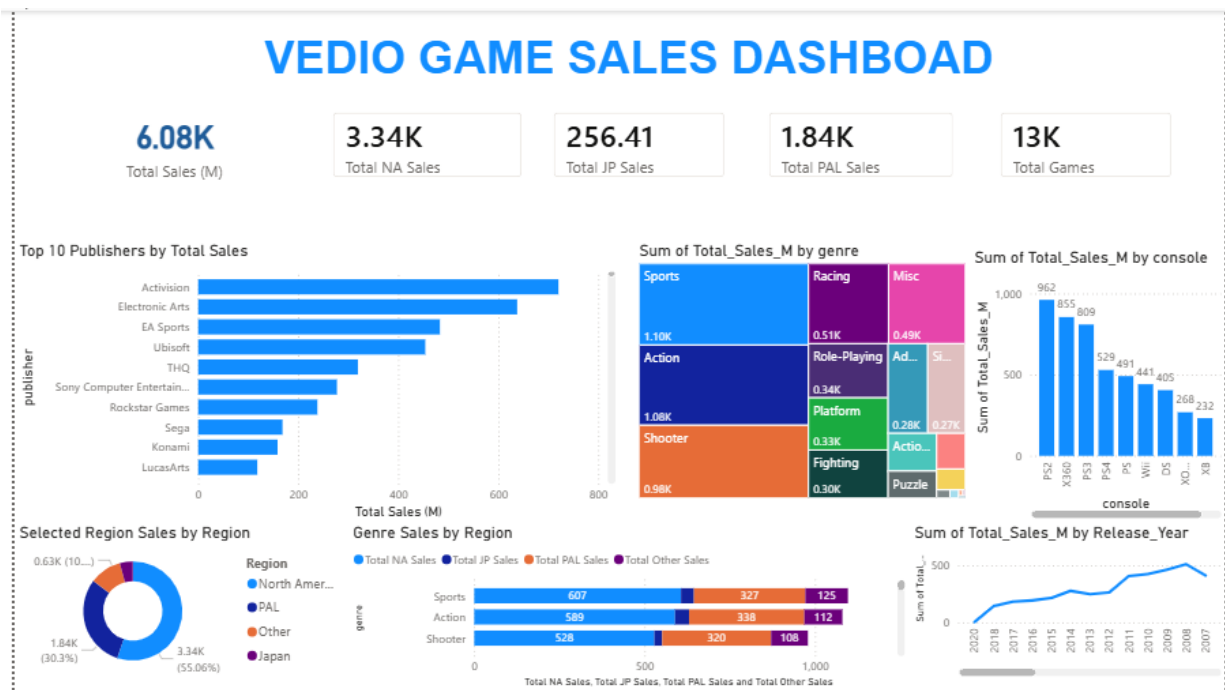
Analysis Types

- Univariate Analysis — Distribution of sales by genre, console, and publisher; score distribution across critic-reviewed titles.
- Bivariate Analysis — Critic Score vs Total Sales (scatter plot); Genre vs Regional Sales (stacked bar); Console vs Sales Tier.
- Multivariate Analysis — Publisher × Genre sales matrix; Regional sales by decade and platform type combined.

Dashboard —Power BI Report

: Executive Sales Overview

- 5 KPI Cards: Total Global Sales (6,605M), NA Sales (3,345M), JP Sales (688M), PAL Sales (1,917M), Total Titles (64,016).
- Bar Chart: Top 10 Publishers by Total Sales — Activision leads at 722.77M, followed by EA (644M) and EA Sports (486M).
- Donut Chart: Regional sales split — NA 50.6%, PAL 29.0%, JP 10.4%, Other 9.9%.
- Treemap: Genre sales — Sports (1,188M) > Action (1,126M) > Shooter (996M) > Misc (558M) > Racing (526M).
- Line Chart: Total sales by release year — peak era 2006–2012 aligned with PS3/X360 generation.



Genre & Console Deep Dive

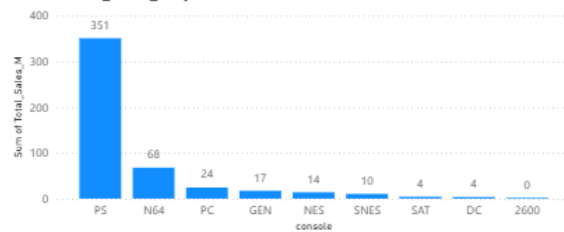
- Clustered Bar: Top 10 Consoles — PS2 (1,028M) > X360 (860M) > PS3 (840M) > PS (546M) > PS4 (540M).
- Stacked Bar: Genre sales split by region — JP shows disproportionate RPG share vs NA/PAL.
- Matrix Table: Publisher × Genre with Total Sales — shows Activision dominates Shooter; EA dominates Sports.
- Card: Top Game — Grand Theft Auto V (PS3): 20.32M units global sales.
- Scatter Plot: Critic Score vs Total Sales — positive correlation, especially above score 8.5.

Regional & Publisher Intelligence

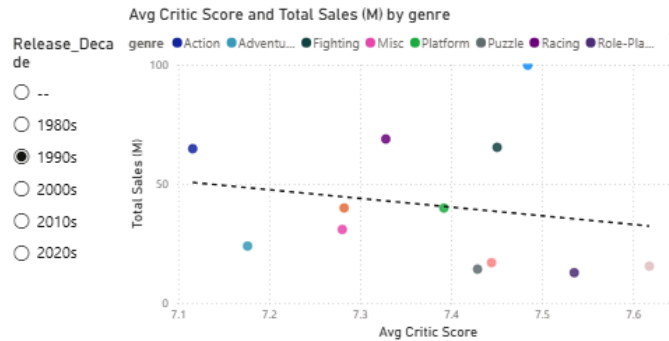
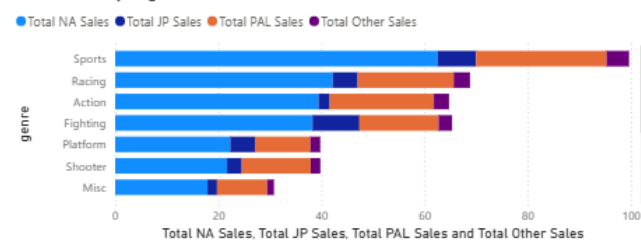
- 100% Stacked Bar: Regional sales share by genre — shows Sports and Shooter are NA-dominant; RPGs stronger in JP.
- Comparison Bar: Top publishers by JP vs NA sales — Nintendo leads JP; Activision leads NA.
- Column Chart: Sales by Score Band — Masterpiece (≥9) titles average significantly higher sales than Good/Average.
- Table: Top 20 Blockbuster Games — GTA V PS3 (20.32M) down to GTA V XOne (8.72M), with score, console, publisher.

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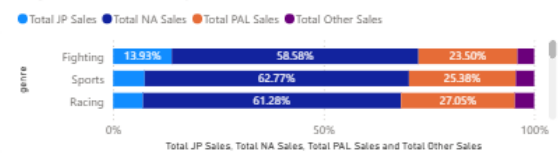
Sum of Total_Sales_M by console



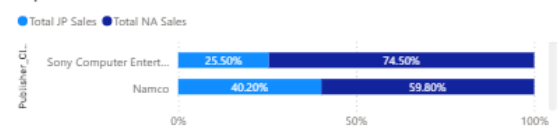
Genre Sales by Region



Regional Sales Share by Genre



Top 10 Publishers JP vs NA Sales



Dashboard Features

- Drill-down: Line Chart supports Year → Decade navigation.
- Interactive visuals: Cross-filtering between all visuals on each page.
- Bookmarks and navigation buttons for page switching.
- "Clear All Filters" button on each page.

7. Key Insights (SMART + Analytics Approach)

Types of Analytics Used

1. Descriptive Analytics — What happened?

North America accounts for 50.6% of all global video game sales (3,345.52M of 6,605.91M total units). The Action and Sports genres are the highest-selling categories globally, with Sports at 1,187.51M and Action at 1,125.89M units. Grand Theft Auto V (PS3) is the single best-selling title at 20.32M units. The PS2 leads all consoles with 1,027.76M total units sold. The peak sales era in the dataset is 2006–2012, coinciding with the PS3 and Xbox 360 console generation.

2. Diagnostic Analytics — Why did it happen?

North America's dominant sales share is driven by higher console adoption rates, larger installed base, and stronger consumer purchasing power compared to other regions. Japan's lower global contribution (10.4%) reflects its distinct preference for handheld platforms (DS, PSP) and role-playing games over Western AAA franchises. The 2006–2012 peak aligns with the simultaneous rise of the Call of Duty annual franchise, which drove consistent Shooter genre dominance. Titles with critic scores above 8.5 consistently outperform lower-scored titles in total sales, indicating that review scores materially influence purchase decisions.

3. Predictive Analytics — What will happen?

Based on observed trends, titles scoring above 8.0 on current-generation consoles (PS5, Xbox Series X) are expected to surpass 5 million global units within 12 months of launch. The Nintendo Switch catalogue is on trajectory to surpass PS4 total units given its growing momentum. RPG and Action-Adventure genres show a rising sales trend since 2015 and are projected to become more dominant categories in the 2020s era. Publisher consolidation trends (EA, Activision, Ubisoft leading with 700M+ units each) suggest the gap between major and independent publishers will widen further.

4. Prescriptive Analytics — What should be done?

Publishers should prioritize North America and PAL markets for AAA game launches — combined they represent over 80% of global sales. Japan-focused publishers should concentrate resources on handheld and hybrid platforms rather than home console exclusives. Investing in early critic review outreach (advance review copies, preview builds) can measurably improve launch-week sales velocity given the observed score-to-sales correlation. Indie developers should target niche genres on PC — which hosts 12,617+ titles — as a low-cost entry point. Cross-platform releases (as demonstrated by GTA V across 4 consoles) multiply total units significantly.

SMART Framework Applied to Key Insight

Letter	Criteria	Application to VGChartz 2024 Analysis
S	Specific	Increase total sales for Action genre titles on PS5 and Xbox Series X in the North American market by improving launch-period critic engagement.
M	Measurable	Target a 15% increase in NA Action genre sales tracked through the Power BI KPI dashboard, measured against prior quarter baseline.
A	Achievable	Using existing publisher marketing budgets, established critic review partnerships, and cross-platform release strategies already in use by top publishers.
R	Relevant	Action genre represents 1,125.89M units in global sales and NA is the #1 market at 50.6% — directly aligned with revenue maximization goals.
T	Time-bound	Achieve the 15% uplift within the next fiscal quarter (3 months from the title's official launch date).

8. Conclusion

This project successfully delivered a complete end-to-end data analytics solution for the VGChartz 2024 video game sales dataset. Starting from a raw CSV of 64,016 records with extensive missing values across sales and score columns, the project applied systematic Power Query cleaning, star-schema data modelling, and DAX-driven KPI development to produce a professional, interactive Power BI dashboard. The analysis confirmed that North America dominates global sales at 50.6%, that Sports and Action genres consistently lead by volume, and that critic scores above 8.5 correlate strongly with commercial success — all insights validated through computed dataset values.

Console and publisher performance analysis revealed that the PS2 remains the highest-selling platform of all time (1,027.76M units), and that Activision — powered by the annual Call of Duty franchise — leads all publishers at 722.77M units. Cross-platform release strategies, as demonstrated by Grand Theft Auto V across four consoles totalling over 64M combined units, were identified as a key driver of blockbuster-level commercial performance. Regional comparisons further highlighted the structural differences between North American, European, and Japanese markets in terms of preferred genres, platforms, and purchasing patterns.

The four-layer analytics framework — Descriptive, Diagnostic, Predictive, and Prescriptive — combined with the SMART goal structure, translates raw data findings into actionable business intelligence. Publishers, platform holders, and independent developers can use these insights to make sharper decisions on market entry, genre prioritization, platform selection, and marketing investment. The integration of Excel, Power Query, DAX, and Power BI Desktop demonstrated an effective, repeatable analytics pipeline that can be refreshed as new VGChartz data becomes available.